

Skills Summary

Writing Growth and Decay Models From Data

Use this reference while completing your worksheet or when studying.

Every model on this sheet has the same shape: $y = a \cdot b^t$, where a is the amount at $t = 0$ and b is what the amount is multiplied by in one time period.

- $b > 1$ grows, $0 < b < 1$ decays, and b is never negative.
- **Writing a model is finding two numbers:** a from the data at time zero, b from a ratio or from a percent rate.
- Say what t counts and what y measures, with its unit. A model without units answers no question.
- Exponential data has a constant *ratio*; linear data has a constant *difference*. Test the ratio before you assume.

1. Reading the factor out of a table

Divide each output by the one before it — later $\div$ earlier. If every quotient is the same number, that number is b . The output at $t = 0$ is a . If the table has no $t = 0$ row, see section 3.

Example: t : 0, 1, 2 with values 24, 36, 54. Write the model.

Step 1. The values are equally spaced in time, so test the ratios: a constant ratio is exactly what an exponential model has.

Step 2. $36 \div 24$ gives the same quotient as $54 \div 36$, so the factor is b , and $a = 24$ is the value at $t = 0$, giving $y = 24(1.5)^t$.

$$b = 1.5$$

Try it: t : 0, 1, 2 with values 5, 20, 80. Find b and write the model.

check: $5(4)^t = 20$ so $4 = b$

2. Turning a percent rate into a factor

Growth of r per period: $b = 1 + r$. Decay of r per period: $b = 1 - r$. Write the percent as a decimal first: 20 percent decay gives $b = 1 - 0.20 = 0.8$. The factor already contains the original amount, so never add the percent on afterwards.

Example: A bike costs 600 dollars and loses 20 percent of its value a year. What is it worth after 3 years?

Step 1. A percent loss each year is a constant multiplier, so build the factor first and let the exponent count the years.

Step 2. Keeping 80 percent gives $b = 0.8$ and $V = 600(0.8)^t$; at $t = 3$ this is $600(0.512)$.

$$307.20$$

Try it: A town of 5000 people grows 6 percent a year. Write the model and find the population after 2 years.

check: $P = 5000(1.06)^t$, giving 5618

3. Two readings, no starting value

If the readings are n periods apart, the factor is applied n times: $(\text{later}) = (\text{earlier}) \cdot b^n$. Solve for b^n , then take the n th root. Keep the positive root only. To get a afterwards, divide any reading by b raised to that reading's own time.

Example: An account holds 800 dollars at $t = 0$ and 1250 dollars at $t = 2$ years. Find the annual factor.

Step 1. The two readings are 2 years apart, so b has been applied twice — that makes the ratio a square, not the factor itself.

Step 2. From $800b^2 = 1250$ comes $b^2 = 1.5625$, and the positive square root of 1.5625 is the annual factor.

$$b = 1.25$$

Try it: A car is worth 1000 dollars at $t = 0$ and 640 dollars at $t = 2$ years. Find the annual factor.

check: $P = 1000(0.8)^t$, so $800 = P$

Watch out. Dividing the *earlier* value by the later one inverts the factor and turns decay into growth — always $\text{later} \div \text{earlier}$. And a constant difference is not a constant ratio: adding 3 percent of the starting amount every year is linear, and always predicts less than the exponential model it was mistaken for.